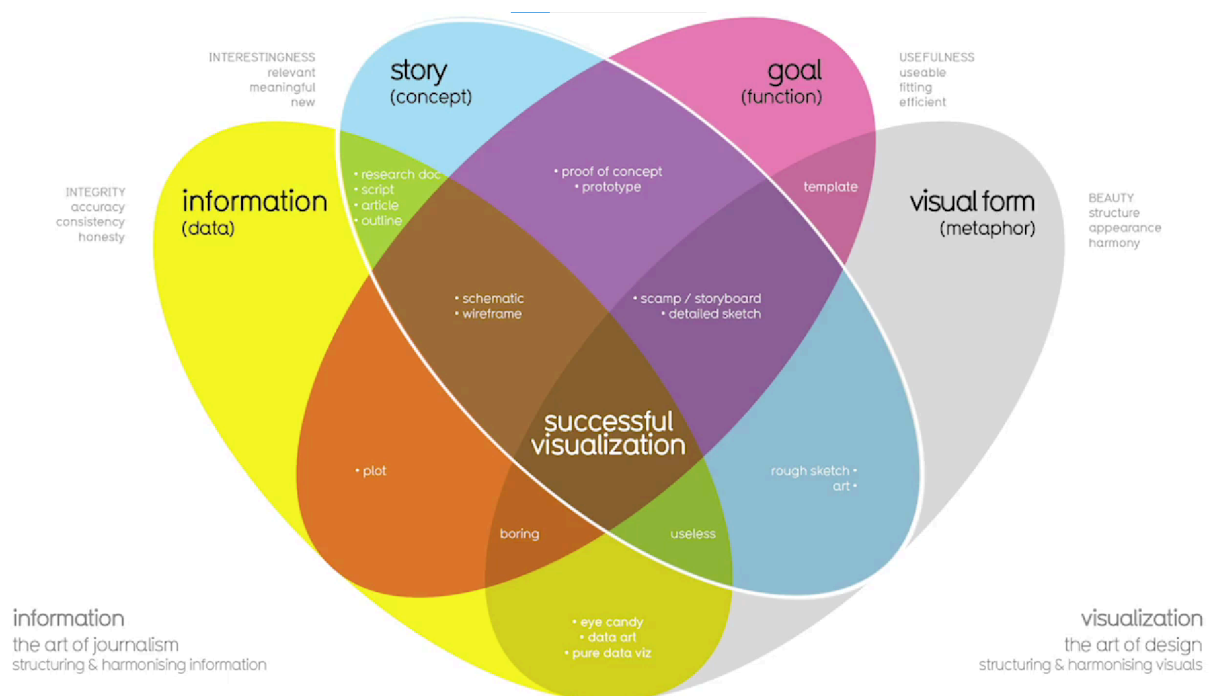




Why data visualization matters



Welcome back, future data analyst. As a budding analyst, you'll be exposed to a lot of data. People learn and absorb data in so many different ways, and one of the most effective ways that this can happen is through visualization. Data visualization is the graphic representation and presentation of data. In reality, it's just putting information into an image to make it easier for other people to understand. If you've ever looked at any kind of map, whether it's paper or online, then you know exactly how helpful visuals could be. Data visualizations are definitely having a moment right now.

Online we are surrounded by images that show information in all kinds of ways, but the history of data visualization goes back way further than the Web. Visualizing data began long ago with maps, which are the visual representation of geographic data. This map of the known world is from 1502. Map makers continued to improve their visualizations as new lands were charted. New data was collected about those locations, and new methods for visualizing the data were created. Scientists and mathematicians began to truly embrace the idea of arranging data visually in the 1700s and 1800s. This bar graph is from 1821 and it doesn't look too different from bar graphs that we see today.

But since the beginning of the digital age of data analytics in the 1990s, the scope and reach of visualizations have grown along with the data they graphically represent. As we keep learning how to more efficiently communicate with visuals, the quality of our insights continue to grow too. Today we can quantify human behavior through data, and we've learned to use computers to collect, analyze and visualize that data. As an analyst in today's world, you'll probably split your time with data visuals in two ways: looking at visuals in order to understand and draw conclusions about data or creating visuals from raw data to tell a story. Either way, it's always good to keep in mind that data visualizations will be your key to success. This is especially true once you reach the point where you're ready to present the results of your data analysis to an audience. Getting people to understand your vision and thought process can feel challenging. But a well-made data visualization has the power to change people's minds. Plus, it can help someone who doesn't have the same technical background or experience as you form their own opinions. So here's a quick rule for creating a visualization. Your audience should know exactly what they're looking at within the first five seconds of seeing it. Basically, this means the visual should be clear and easy to follow. In the five seconds after that, your audience should understand the conclusion your visualization is making. Even if they aren't totally familiar with the research you've been doing. They might not agree with your conclusion, and that's okay. You can always use their feedback to adjust your visualization and go back to the data to do further analysis. So now let's talk about what we have to do to create a visualization that's understandable, effective and, most importantly, convincing. Let's start from the beginning. Data visualizations are a helpful tool for fitting a lot of information into a small space. To do

this, you first need to structure and organize your thoughts. Think about your objectives and the conclusions you've reached after sorting through data.

Then think about the patterns you've noticed in the data, the things that surprised you and, of course, how all of this fits together into your analysis. Identifying the key elements of your findings help set the stage for how you should organize your presentation. Check out this data visualization made by David McCandless, a well-known data journalist. This graphic includes four key elements: the information or data, the story, the goal and the visual form. It's arranged in a four-part Venn diagram, which tells us that all four elements are needed for a successful visualization. So far, you've learned a lot about the data used in visualizations. That's important because it's a key building block for your visualization.

The story or concept adds meaning to the data and makes it interesting. We'll talk more about the importance of data storytelling later, but for now, just remember that the story and the data combined provide an outline of what you're trying to show. The goal or function makes the data both useful and usable, and the visual form creates both beauty and structure. With just two elements, you can create a rough sketch of a visual. This could work if you're at an early stage, but won't give you a complete visualization because you'd be missing other key elements. Even using three elements gets you closer, but you're not quite finished. For example, if you combine information, goal, and visual form without any story, your visual will probably look fine, but it won't be interesting. On their own, each element has value, but visualizations only become truly powerful and effective when you combine all four elements in a way that makes sense. And when you think about all of these elements together, you can create something meaningful for your audience.

At Google I make sure to develop visualizations to tell stories about data that include all four of these elements, and I can tell you that each element is a key to a visualization success. That's why it's so important for you as the analyst to pay close attention to each element as we move forward. Other people might not know or understand the exact steps you took to come to the conclusions you've made, but that shouldn't stop them from understanding your reasoning. Basically, an effective data visualization should lead viewers to reach the same conclusion you did, but much more quickly.

Because of the age we live in, we're constantly being shown different ways to view and absorb information. This means that you've already seen lots of visuals you can reference as you design your own visualizations. You have the power to tell convincing stories that could change opinions and shift mindsets.

That's pretty cool. But you also have the responsibility to pay attention to the perspectives of others as you create these stories. So it's important to always keep that in mind. Coming up, we'll start drawing connections between data and images to create a strong foundation for your visual masterpieces. I can't wait to get started.